

KSHITIZ KHANDELWAL

Bangalore, India | +91 8880220080 | kshitiz.k.1403@gmail.com | github.com/Kshitiz-Khandelwal |
linkedin.com/in/kshitiz-khandelwal-94917b32a

EDUCATION

BMS Institute of Technology & Management — *B.E. Computer Science and Business Systems* **2024 – 2028**

CGPA: **9.43** | *Relevant Coursework: Data Structures, Operating Systems, Machine Learning, DBMS, Embedded IoT* Bangalore, India

TECHNICAL SKILLS

Languages & Systems: Python, C, C++, Rust, TypeScript, JavaScript, SQL, Bash
AI/ML & Vision: PyTorch, LightGBM, XGBoost, Scikit-Learn, Continuous Wavelet Transform (CWT), Grad-CAM, TreeSHAP
Full Stack & Edge: Next.js 15, FastAPI, React, Node.js, Tailwind CSS, Zephyr RTOS, Raspberry Pi Pico 2 (RP2350), ESP32, Docker

KEY ENGINEERING & RESEARCH PROJECTS

DNS Shield AI Platform — High-Throughput Threat Detection *Python, FastAPI, LightGBM, Next.js*

- Engineered an AI threat filter classifying 100K+ queries/sec with Shannon entropy extraction and DGA attribution.
- Integrated real-time TreeSHAP feature attribution explaining domain lexical decisions with under 1.2ms inference latency.

Privacy-Preserving Federated Healthcare (SplitFed) *PyTorch, Federated ML, FastAPI*

- Built decentralized multi-hospital image diagnosis using cut-layer feature smashes and NoPeek loss, preserving HIPAA data privacy.
- Achieved 94.8% multi-class pathology classification without transferring raw patient radiographs to central servers.

12-Lead ECG Arrhythmia Classification via Continuous Wavelet Transforms *PyTorch, CWT, EfficientNet, SHAP*

- Converted 12-lead raw biosignals into 2D time-frequency scalograms using Morlet wavelet transforms for cardiac anomaly detection.
- Trained EfficientNet-B4 reaching 97.4% F1-score across 5 arrhythmia classes with Grad-CAM visual heatmaps.

Zephyr RTOS Edge AI & IoT Lab (RP2350 Pico 2) *C/C++, Zephyr RTOS, West, ARM Cortex-M33*

- Developed real-time multi-threaded firmware on dual-core RP2350 with deterministic sensor sampling and edge telemetry.

LEADERSHIP & HACKATHONS

Team Lead — Krishi360 Platform (College Ideathon, SIH-Aligned PS) *3rd Place Podium*

- Led 6-developer squad architecting Crop Suitability Index mathematical model and live advisory platform.

Co-Organizer & Instructor — IoT & Edge AI Technical Outreach Workshop *May 2026*

- Instructed 100+ students from rural engineering colleges in microcontroller programming, sensors, and cloud telemetry.